به نام خدا

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پاسخ تمرین جلسه چهارم

برنامه ای بنویسید که شماره متناظر با کلید فشرده شده در کی پد را در LCD به صورت پشت سر هم نمایش دهد.

G:\Personal\Uni\Micro Lab\Micro\Az4-Tamrin\Az4-Tamrin.c

Notes | Az4-Tamrin.c

```
/*****************************************************
This program was created by the
CodeWizardAVR V3.12 Advanced
Automatic Program Generator
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Project : Session4-Exercise
Version :
Date    : 10/17/2020
Author  : Ali Sepehrnia (96121038)
Company :
Comments:


Chip type               : ATmega16
Program type            : Application
AVR Core Clock frequency: 1.000000 MHz
Memory model            : Small
External RAM size       : 0
Data Stack size         : 256
*****************************************************/

#include <mega16.h>
#include <delay.h>

// Alphanumeric LCD functions
#include <alcd.h>

char ReadKeypad()
{
    PORTC = 0xff;   //1111(pull up) 1111(scan)

    //--Row 1--//
    PORTC.4 = 0;
```

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```
    //--Row 1--//
    PORTC.4 = 0;
    if(PINC.0 == 0) return '7';
    else if(PINC.1 == 0) return '8';
    else if(PINC.2 == 0) return '9';
    else if(PINC.3 == 0) return '/';
    PORTC.4 = 1;

    //--Row 2--//
    PORTC.5 = 0;
    if(PINC.0 == 0) return '4';
    else if(PINC.1 == 0) return '5';
    else if(PINC.2 == 0) return '6';
    else if(PINC.3 == 0) return '*';
    PORTC.5 = 1;

    //--Row 3--//
    PORTC.6 = 0;
    if(PINC.0 == 0) return '1';
    else if(PINC.1 == 0) return '2';
    else if(PINC.2 == 0) return '3';
    else if(PINC.3 == 0) return '-';
    PORTC.6 = 1;

    //--Row 4--//
    PORTC.7 = 0;
    if(PINC.0 == 0) return 'c';
    else if(PINC.1 == 0) return '0';
    else if(PINC.2 == 0) return '=';
    else if(PINC.3 == 0) return '+';
    PORTC.7 = 1;

    return '';
}

// Declare your global variables here
```

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```
// Declare your global variables here

void main(void)
{
// Declare your local variables here
char key;

// Input/Output Ports initialization
// Port A initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRA=(0<<DDA7) | (0<<DDA6) | (0<<DDA5) | (0<<DDA4) | (0<<DDA3) | (0<<DDA2) | (0<<DDA1) | (0<<DDA0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTA=(0<<PORTA7) | (0<<PORTA6) | (0<<PORTA5) | (0<<PORTA4) | (0<<PORTA3) | (0<<PORTA2) | (0<<PORTA1) | (0<<PORTA0);

// Port B initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRB=(0<<DDB7) | (0<<DDB6) | (0<<DDB5) | (0<<DDB4) | (0<<DDB3) | (0<<DDB2) | (0<<DDB1) | (0<<DDB0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTB=(0<<PORTB7) | (0<<PORTB6) | (0<<PORTB5) | (0<<PORTB4) | (0<<PORTB3) | (0<<PORTB2) | (0<<PORTB1) | (0<<PORTB0);

// Port C initialization
// Function: Bit7=Out Bit6=Out Bit5=Out Bit4=Out Bit3=In Bit2=In Bit1=In Bit0=In
DDRC=(1<<DDC7) | (1<<DDC6) | (1<<DDC5) | (1<<DDC4) | (0<<DDC3) | (0<<DDC2) | (0<<DDC1) | (0<<DDC0);
// State: Bit7=0 Bit6=0 Bit5=0 Bit4=0 Bit3=P Bit2=P Bit1=P Bit0=P
PORTC=(0<<PORTC7) | (0<<PORTC6) | (0<<PORTC5) | (0<<PORTC4) | (1<<PORTC3) | (1<<PORTC2) | (1<<PORTC1) | (1<<PORTC0);

// Port D initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRD=(0<<DDD7) | (0<<DDD6) | (0<<DDD5) | (0<<DDD4) | (0<<DDD3) | (0<<DDD2) | (0<<DDD1) | (0<<DDD0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTD=(0<<PORTD7) | (0<<PORTD6) | (0<<PORTD5) | (0<<PORTD4) | (0<<PORTD3) | (0<<PORTD2) | (0<<PORTD1) | (0<<PORTD0);

// Alphanumeric LCD initialization
// Connections are specified in the
// Project|Configure|C Compiler|Libraries|Alphanumeric LCD menu:
// RS - PORTA Bit 0
```

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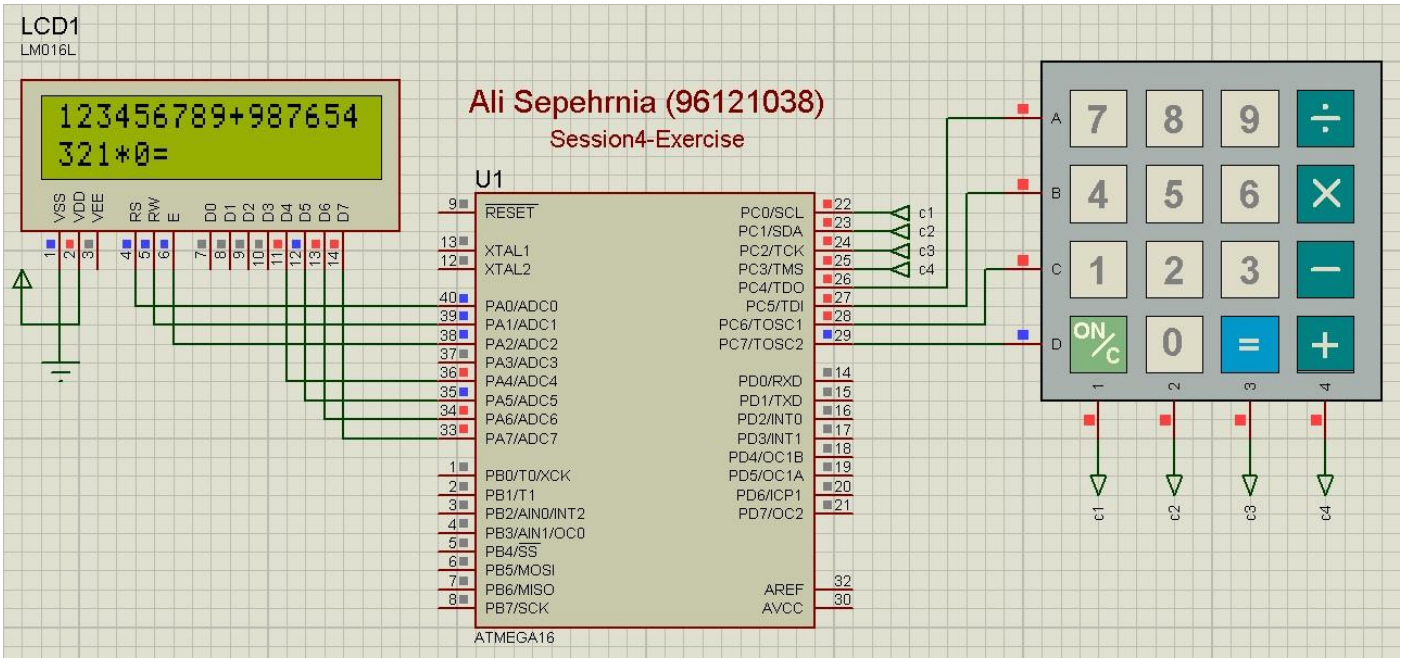
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```
// RS - PORTA Bit 0
// RD - PORTA Bit 1
// EN - PORTA Bit 2
// D4 - PORTA Bit 4
// D5 - PORTA Bit 5
// D6 - PORTA Bit 6
// D7 - PORTA Bit 7
// Characters/line: 16
lcd_init(16);
lcd_clear();
lcd_gotoxy(0,0);

while (1)
      {
      // Place your code here
      do
            key = ReadKeypad();
        while(key == '');

      if(key=='c')  //Clear screen if On/C key pressed
        lcd_clear();
      else
        lcd_putchar(key);

      delay_ms(250);
      }
}
```


LCD1
LM016L
123456789+987654
321*0=
Ali Sepehrnia (96121038)
Session4-Exercise
U1
ATMEGA16
RESET
XTAL1
XTAL2
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PC0/SCL
PC1/SDA
PC2/TCK
PC3/TMS
PC4/TDO
PC5/TDI
PC6/TOSC1
PC7/TOSC2
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
AREF
AVCC
c1
c2
c3
c4
ON/C